

POWER OASIS™ MEDICAL

Isolated power that moves with your OR.

Portable power station designed to NFPA 99 Category 1* isolated power requirements — hospital-grade isolation transformer, line isolation monitoring, and online UPS in a single cart.

- **NFPA 99 Category 1*** isolated power architecture[4]
- UL 60601-1* isolation transformer
- **Line Isolation Monitor** — alarm at ≤5mA hazard current
- **Online UPS** — no transfer time
- **3h 27m @ 1,000W**

48	100	4800	~1,440	6	3h 27m
V	Ah	Wh	W		@ 1,000 W
SYSTEM VOLTAGE	CAPACITY	STORED ENERGY	ISOLATED OUTPUT	120V AC OUTLETS	OR RUNTIME

OR RUNTIME AT LOAD

Coverage for the clinical scenarios that matter

OR Load	Runtime	Clinical Context
500W	6h 54m	Anesthesia monitor · infusion pump · lighting
750W	4h 36m	Multi-device light OR load
1,000W	3h 27m	Typical mixed OR — anesthesia + monitor + ESU
~1,440W	2h 24m	Rated continuous maximum

At 80% DoD · 90% combined efficiency · 100Ah battery. Lightweight 60Ah configuration planned (~30 lbs · 2h 04m @ 1,000W). Designed as backup for unexpected outages — recharges in ~6.7 hrs.

KEY SPECIFICATIONS

Energy	**4,800 / 3,456 Wh** Gross / usable AC	**Recharge**	**~6.7 hrs** 15A DC integrated charger
Inverter	**48/3000** Online UPS	**Outlets**	**6 positions · on inline transformer** Hospital-grade outlets · ~1,440 W total across all outlets
Isolation transformer	**~1,440 W continuous** UL 60601-1 3rd ed*	**AC input**	**15A DC charger · HG plug**
Output leakage	**<100 µA** Faraday shield · NFPA 99* limit	**EMI filter**	**medical** <2µA leakage — B-type
Line isolation monitor	**≤5mA alarm** Visual + audible · front panel[5]	**Enclosure**	**5052-H32 aluminum** wipe-clean
Battery	**48V 100Ah LiFePO4** ~75 lbs[2] · 60Ah lightweight option	**Weight**	**~214 lbs[2]** Verify facility placement policy

*[2] Estimated — pending vendor confirmation. [4] Compliance path under regulatory review — see notices. [5] LIM make/model selection in progress. * Future certification — testing and planning phase.*

ISOLATED POWER ARCHITECTURE — NFPA 99 CATEGORY 1*

Why each element exists and what it protects

Element	Component	Function
Isolation transformer	Hospital-grade transformer	Breaks ground reference
Line isolation monitor	LIM (model TBD[5]) · front panel	Monitors isolation integrity · alarms ≤5mA · powered from isolated output side per NFPA 99*
Output leakage	<100µA	Transformer Faraday shield
No GFCI on OR outlets	Intentional design	GFCI would cut ALL OR power on first fault — defeats isolated power purpose
Online UPS	Online inverter/charger	Inverter runs continuously — no transfer time
Fuse monitoring	DC distribution + CAN bus	Blown DC fuse → visible system display alarm

ADDITIONAL SPECIFICATIONS

Complementing key specifications on page 1

Rated output	**3,000 VA · 5,000 W surge** inverter/charger	**Transformer weight**	**58 lbs** Two-person install lift
Max discharge	**0.33C** At ~1,440 W · 67% headroom to 1C	**Cooling**	**2× 120mm fans**
Charge voltages	**54.75 / 52.0 V DC** LiFePO4 absorption / float	**Input plug**	**Hospital-grade plug**
Efficiency	**~93% inverter · ~97% transformer** Combined ~90%	**Operating temp**	**−10 to +40 °C** Hospital environment

WARRANTY & SUPPORT

Coverage	Details
Product warranty	2 years parts/labor (target) · 5 years battery and power electronics
Annual service	LIM verification · leakage measurement · battery health — included year 1
Remote diagnostics	Cloud monitoring — biomedical team and our support see identical live data
Compliance docs	Complete NFPA 99* documentation package planned

⚠ IMPORTANT NOTICES

- CONCEPT STAGE — NOT for clinical use. NFPA 99* compliance path under regulatory review; deployment pending confirmation.
- EMI filter is medical B-type exclusively — standard filters violate leakage limits.
- No GFCI on OR outlets is intentional — see architecture above.

* Future certification — testing and planning phase.

Planning an OR renovation or resilience program?

Concept Stage — specifications preliminary and subject to change — not released for production sales@trifectaengineering.com